MIL-B-36893A
21 February 1975
Superseding
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7 November 1972

MILITARY SPECIFICATION

BRONCHOSCOPE, FIBER OPTIC, HOLINGER

This specification is approved for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fiber optic bronchoscopes, of the Holinger design, suitable for use in bronchoscopy and anesthesiology.

1.2 Classification. Bronchoscopes shall be furnished in the following sizes, as indicated below (see 6.1):

Size	
1	3.5 mm Diameter lumen by 25 cm long.
2	5 mm Diameter lumen by 30 cm long.
3	7 mm Diameter lumen by 40 cm long.
4	9 mm Diameter lumen by 40 cm long.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. These documents may be obtained as directed by the contracting officer.

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-116	Preservation, Methods of.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24304	Bronchoscope, Fiber Optic, Holinger

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI Z79.2 Endotracheal-Tube Connectors and Adapters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. The main tube, light carrier channel and light carrier housing shall be fabricated from unplated nickel-copper alloy (Monel). As an alternate, the distal tip of the main tube shall be fabricated from class 8, type 752 material in accordance with Federal Specification GG-I-526, except that the material shall not be plated. The handle, entire air tube assembly, bayonet connection, window plug housing, cap and spacers shall be fabricated from class 3, type 302, 303, or 304 material in accordance with GG-I-526. The window of the window plug assembly shall be fabricated from clear polycarbonate. The sealing members of the window plug assembly shall be fabricated from polytetrafluoroethylene (Teflon). The light carrier shall be filled with suitable glass fibers to disperse illumination.

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3.2 Style, design and dimensions. Shall be a fiber optic bronchoscope, of the Holinger design, suitable for use in bronchoscopy and anesthesiology when used with National Stock Numbers 6515-00-104-8520 LIGHT, ENDOSCOPIC INSTRUMENT, FIBER OPTIC, 115 Volt, 60 Hz, AC and 6515-00-104-8434 CABLE, FIBER OTPIC, ENDOSCOPTIC INSTRUMENT, 72 inches long. The instrument shall be furnished with an oval-shaped fiber optic light carrier that will disperse illumination when assembled into the light carrier channel of the instrument and assembled to the national stock numbers specified above. The instrument shall have a smooth distal tip to provide maximum visibility and ease of instrument insertion. The instrument shall have an air tube assembly, with a 15 millimeter (mm) endotracheal connector permanently secured to the air tube. The air tube shall be permanently secured to the main tube proper. The 15 mm connector shall have a standard external taper in accordance with Figure 2 of ANSI Standard Z79.2 to accept standard endotracheal tubes for anesthesiology and aspiration procedures. The handle of the bronchoscope shall be permanently secured to the main tube and possess an external taper at the proximal end which will accommodate an interchangeable window plug to provide a closed system for administering anesthesia. The fiber optic light carrier shall be of the bayonet type that shall be readily inserted into the light carrier channel and positioned into place by a slot in the instrument handle. The light carrier channel shall be permanently secured to the main tube and shall be completely sealed at the outer surface of the distal end. Open slots shall be on both sides of the main tube at the distal end. When the light carrier is inserted into the light carrier channel, the lumen of the instrument shall not be decreased in size. The diameter of the lumen (as defined on the drawing) shall be as follows for the indicated sizes:

Size	Diameter (in millimeters, mm)
1	3.5
2	5
3	7
4	9

The bronchoscopes shall be in accordance with DPSC Drawing Number 24304 for the appropriate size and national stock number.

3.3 Finish. The complete instrument, including all components, shall have a smooth finish throughout. Burrs, cracks, holes, pits, corrosion, and sharp edges shall be prohibited. The distal edge of the instrument shall be smoothly blended to the surfaces of the main tube.

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3.4 Corrosion resistance. All components shall be subjected to the boil test as specified in GG-I-526. In addition, all components fabricated from nickel-copper alloy or class 8 material shall be subjected to the salt-spray test specified in GG-I-526.

3.5 Instrument operation. When tested as specified in 4.4.1, the light carrier shall be capable of adequate insertion into the light carrier channel. Illumination shall be readily observed at the distal end of the light carrier. Failure of the window plug to form a tight fit on the external taper portion of the handle shall also be cause for rejection.

3.6 Identification marking. The instrument shall be permanently marked with the name or registered trademark of the manufacturer, size, and country of origin. Further, the light carrier shall bear permanent, appropriate, length identification marking in centimeters.

3.7 Container. Each instrument shall be furnished in a suitable container, compartmented and cushioned, to prevent damage during transit.

3.8 Workmanship. Workmanship shall be first class throughout. The instruments shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the nickel-copper alloy, nickel-silver alloy and corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one complete bronchoscope.

Table I. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
Critical	100 percent	No defects permitted
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for testing purposes shall be one complete bronchoscope. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Bronchoscope	Dimensions	3.2	----	S-1	1.5
	Boil test (all material)	3.4	GG-I-526	S-1	1.0
	Salt-spray test (class 8 and nickel-copper alloy)	3.4	GG-I-526	S-1	1.0
	Operation test	3.4	4.4.1	100%	*

* No defects permitted.

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4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the bronchoscopes shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of bronchoscopes. The bronchoscopes shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any instruments in the sample containing one or more defects shall be rejected and if the number of defective instruments in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	
01	Outer surfaces of instruments proper not free from sharp edges and burrs.
02	Handle, light carrier channel, and air tube assembly (including adapter) not permanently secured to main tube.
Major	
101	Bronchoscope not of the specified design and size.
102	Light carrier and window plug not furnished with unit.
103	Light carrier not of bayonet design.
104	Glass fibers not completely encased and sealed in light carrier housing.
105	Lumen of instrument not free from foreign matter.
106	Finish of lumen (including window plug) not free from cracks, holes, pits, burrs, sharp edges and corrosion. Finish of outer surfaces not free from cracks, holes, pits and corrosion.

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Table III. Classification of Defects (Cont'd)

Categories	Defects
107	Lumen obstructed when light carrier is assembled to light carrier channel.
108	Instrument plated.
109	Light carrier channel not completely sealed at outer surface of distal end.
110	Distal end not smoothly contoured to main tube.
111	Size markings not present.
112	Window broken, cracked, crazed and not clear.
113	Air tube obstructed or occluded.
Minor	
201	Bronchoscope not identified as specified in 3.6.

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.

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Table IV. Examination for Preparation for Delivery (Cont'd)

Examination	Defect
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, taper improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operation test. This test shall be performed on a 100 percent basis. Insert the light carrier into the light carrier channel. Position the bayonet handle into the slot of the instrument stem. Place the window plug onto the external taper portion of the handle. Assemble fiber optic cable, with bayonet connection (NSN 6515-00-104-8434) to bayonet portion of light carrier. Assemble fiber optic cable to endoscopic light source (NSN 6515-00-104-8520). With the light source properly connected to a 115 volt, 60 Hz, AC power source, observe for illumination of the light carrier by viewing through the window plug. Illumination shall not flicker during the test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each bronchoscope and components shall be thoroughly cleaned and dried in accordance with applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit package. Each bronchoscope and components, packaged as specified in 3.7, shall be further packaged in accordance with MIL-P-116, method 1C-1.

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5.1.1.4 Intermediate package. Twelve (12) bronchoscopes shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, Type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Forty eight (48) bronchoscopes of sizes 1 or 2; or twenty four (24) bronchoscopes of sizes 3 or 4 shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. Forty eight (48) bronchoscope of sizes 1 or 2; or twenty four (24) bronchoscopes of sizes 3 or 4 shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-104-9318	BRONCHOSCOPE, Fiber Optic, Holinger, 7 mm by 40 cm.
6515-00-890-3512	BRONCHOSCOPE, Fiber Optic, Holinger, 5 mm by 30 cm.
6515-00-890-3518	BRONCHOSCOPE, Fiber Optic, Holinger, 3.5 mm by 25 cm.
6515-00-890-3522	BRONCHOSCOPE, Fiber Optic, Holinger, 9 mm by 40 cm.

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1928

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-B-36893A BRONCHOSCOPE, FIBER OPTIC, HOLINGER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

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